

Dexter Fernandes

Robotics Engineer

Bristol, UK (open to relocating to London) | +44 7436 007369 | dexterfernandes11@gmail.com

[linkedin.com/in/dexterferns](https://www.linkedin.com/in/dexterferns) | github.com/Dexter-Fernandes

PROFILE

Robotics engineer working across perception, state estimation and real-time systems. Production background training, optimising and deploying computer vision on constrained edge hardware for 24/7 industrial operation, with an MSc in Robotics covering LiDAR SLAM, pose-graph and factor-graph optimisation, and multi-sensor calibration. Recent self-directed work spans TensorRT and CUDA plugin development, multi-object tracker benchmarking, and EKF-based sensor fusion and navigation in ROS2 and Gazebo simulation. Comfortable taking an unfamiliar problem from prototype to on-site trials and explaining the result to non-specialists.

EXPERIENCE

Independent Robotics & Perception Engineering — Self-Directed

Sep 2024 – Present

Bristol, UK

- Full-time self-directed engineering programme following the MSc, rebuilding production-grade perception, state estimation and simulation systems end to end. Three projects below; all code and benchmark data published on GitHub.

Multi-Stream Video Analytics Pipeline — DeepStream / TensorRT

Jun 2026

- Built an end-to-end NVIDIA DeepStream 9.0 / GStreamer RTSP video analytics pipeline for multi-stream computer vision, chaining rtspsrc, nvstreammux, nvinfer, nvtracker and nvdsosd to run YOLO detection with NvDCF tracking, metadata extraction, anonymisation, CSV logging and re-streamed output across 3 concurrent camera feeds.
- Deployed and optimised a YOLO26n TensorRT FP16 detector on 6 GB VRAM Turing hardware, combining dynamic-batch engine conversion, nvinfer integration, p99 latency profiling and FP16-vs-FP32 accuracy validation to sustain 29.7 FPS per stream live and 131.3 FPS per stream unthrottled.
- Implemented a C++/CUDA TensorRT decode plugin using IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked multi-object tracker configurations across IOU, Kalman-filter-based NvDCF, and ByteTrack-inspired NvSORT, producing MOTA, IDF1, ID switches, fragmentation, per-frame detection counts, FPS and VRAM impact to guide production tracker selection.
- Hardened for production with GPU smoke tests, model-promotion gates, structured JSON logging and per-camera health metrics covering liveness, dropped frames, reconnect events and RSS.

ROS2 State Estimation, SLAM & Navigation in Simulation

Feb 2026

- Built a simulation-first robotics stack in ROS2 Jazzy and Gazebo Harmonic to develop and test LiDAR-centric SLAM, LiDAR+RGB fusion SLAM and autonomous navigation before committing to hardware.
- Implemented an EKF-based localisation layer with robot_localization, fusing wheel odometry and IMU to maintain a smooth, continuous odom-to-base estimate through LiDAR-only SLAM and degraded-measurement periods.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping accuracy, localisation drift and runtime behaviour to support system selection.
- Developed ROS2 nodes, launch files and parameterised configurations in C++17 and Python, using tf2 frame management, multi-sensor message synchronisation and timestamp alignment to resolve transform timing issues across camera and LiDAR ingestion.
- Implemented point cloud processing with PCL and Open3D for filtering, segmentation and registration, and containerised the stack with Docker for reproducible builds across environments.

Modern Edge CV Model Benchmarking

Mar 2026

- Benchmarked YOLOv8, YOLOv10, YOLO26, RT-DETR, EfficientViT, DINO, MobileSAM and Depth Anything v2 on accuracy, latency, model size and deployment suitability, producing a written comparison to guide detector selection for constrained real-time targets.

Esbaar — AI Engineer

Jan 2023 – Aug 2023

Muscat, Oman

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training, deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on BP drilling rigs.
- Led on-site validation and diagnostics on live deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that measurably improved robustness in operation.

- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries and safe restarts, reducing live video pipeline crashes by 67%.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, giving consistent latency, memory and stability outcomes across deployments.
- Presented performance findings to leadership and multinational stakeholders, from project managers to drill rig operators, translating technical results into deployment decisions.

LivNSense Technologies — Lead Software Engineer, AI

May 2020 – Dec 2022

Bengaluru, India

- Led a team designing, training and deploying vision pipelines for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries.
- Built low-latency asynchronous perception pipelines in C++17 with OpenVINO, increasing throughput from 5 to 15 FPS and delivering deterministic real-time inference on Intel hardware.
- Deployed quantised vision models using FP16/INT8 optimisation, structured pruning and layer fusion, reducing model size by 50% while preserving production accuracy targets.
- Profiled and optimised C++ inference and pre/post-processing hotspots with Intel VTune for targeted performance fixes in the CNN inference pipeline.
- Tuned camera and ISP settings to keep models reliable on RGB+IR streams under 24/7 operation, cutting false positives on IR and reducing manual recalibration.

EDUCATION

MSc Robotics — Merit, University of Bristol

Sep 2023 – Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, fusing LiDAR and radiological sensor data for localisation and mapping in inspection scenarios.
- Configured Ceres, g2o and GTSAM as factor-graph and pose-graph optimisation backends, and tuned ICP parameters across environments.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base for the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, with tf2 frame management and message synchronisation to resolve transform timing issues.

BEng/BTech Electrical & Electronics Engineering, VNIT Nagpur

Jul 2016 – May 2020

TECHNICAL SKILLS

- **State Estimation & Sensor Fusion:** Kalman filtering, EKF (robot_localization), multi-sensor fusion, factor-graph and pose-graph optimisation (GTSAM, g2o, Ceres), ICP, scan matching, LiDAR SLAM (RTAB-Map, LIO-SAM, Cartographer, GLIM), extrinsic calibration, tf2
- **Computer Vision & ML:** Object detection and multi-object tracking (YOLOv8/v10/26, RT-DETR, EfficientDet, NvDCF, NvSORT, ByteTrack), PyTorch, TensorFlow, OpenCV, PCL, Open3D, CVAT, Optuna
- **Real-Time & Edge Deployment:** TensorRT with custom CUDA plugins (IPluginV2DynamicExt), CUDA, NVIDIA DeepStream 6.x/9.0, GStreamer, RTSP pipelines, JetPack, OpenVINO, ONNX, FP16/INT8 quantisation, QAT and PTQ, structured pruning, layer fusion, latency and throughput profiling
- **Robotics & Simulation:** ROS2 (Jazzy, rclcpp), Gazebo Harmonic, Nav2, robot_localization, rosbag2, Velodyne VLP-16, Intel RealSense D435i, RGB+IR camera and ISP tuning, camera behaviour debugging
- **Programming & Tooling:** C++17, Python, CUDA C++, Bash, CMake, Git, Linux, Intel VTune, FFmpeg
- **Systems & MLOps:** Docker, CI/CD, Apache Kafka, MLflow, DVC, Airflow, Microsoft Azure, structured logging, health checks, observability, model-promotion gates, production monitoring